

KRATIKA RATHI

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EDUCATION

University of Southern California

Master of Science in Computer Science (Artificial Intelligence)

Los Angeles, CA

May 2027

Relevant Coursework: Theoretical Machine Learning, Algorithm Design and Analysis, Natural Language Processing

The University of British Columbia

Bachelor of Science in Statistics

Vancouver, BC

May 2025

Relevant Coursework: Bayesian Statistics, Multivariate Statistical Analysis, Probability, Time Series and Forecasting

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, C++, JavaScript

ML/NLP: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy, XGBoost, statsmodels

LLM Development: Ollama, OpenAI API, Anthropic API, LangChain, ChatGPT, Claude

Tools: Git, GitHub, Linux, Docker, Tableau, Power BI, Jira, Cursor

EXPERIENCE

Graduate Researcher

June 2026 - Present

Melady Lab, University of Southern California

Los Angeles, CA

- Designed STRATA, an LLM-agent framework that decomposes agent skills into reusable procedure cores and environment-specific adapters to improve cross-environment skill transfer.
- Developed self-improving agents that revise written instructions from execution feedback and added evaluation safeguards to detect performance regression on previously explored environments.
- Curated benchmark datasets and implemented evaluation pipelines to measure task success, transfer performance, and generalization across environments.

Actuarial Intern

January 2024 - August 2024

iA Financial

Vancouver, BC

- Performed monthly reserve calculations for the Special Markets Group using chain-ladder methods and analyzed reserve adequacy and portfolio performance.
- Conducted profitability studies and applied GARCH modeling to analyze risk and forecast claim patterns, contributing to a 15% improvement in financial forecasting.
- Designed and validated survey instruments that reduced data-entry and validation errors by 10%, and built actuarial pricing, reserving, and valuation models to support risk assessment.

Data Analyst

January 2023 - April 2023

Orennia

Calgary, AB

- Built and maintained multinomial logistic regression models to predict power-plant viability using capacity, location, and size, supporting client investment decisions across U.S. energy markets.
- Automated forecast generation with custom Python scripts, improving the efficiency and reproducibility of recurring modeling workflows.

Data Analyst

May 2022 - December 2022

Animal Welfare Program, University of British Columbia

Vancouver, BC

- Applied mixed-effects models to assess the repeatability and predictability of cattle feeding and drinking behavior for animal-welfare monitoring.
- Labeled and analyzed high-volume behavioral data to identify dominance hierarchies among dairy cattle and support early detection of illness, lameness, and missing animals.
- Optimized Python data-processing pipelines and contributed to statistical analysis, visualization, interpretation, and technical research reporting.

PROJECTS

Fault in Our Stars: Astronomical Light-Curve Classification | GitHub

- Built an end-to-end Python and scikit-learn time-series classification pipeline over 15,523 processed Kepler light curves, distinguishing confirmed exoplanets, eclipsing binaries, variable stars, and false positives.
- Implemented robust preprocessing with MAD-based sigma clipping, median normalization and detrending, KIC-level deduplication, and interpolation of variable-length flux sequences to 300-point feature vectors.
- Trained a 300-tree, class-balanced Random Forest with stratified splitting and SMOTE; evaluated it using macro-F1, per-class precision/recall, and confusion matrices, establishing a 0.57 macro-F1 four-class baseline.

Classifying Mantle Compositions Using Geochemical Data from Hawaiian Lavas

- Built an end-to-end machine-learning pipeline for trace-element geochemical data, including missing-value imputation, feature engineering, and supervised model evaluation.
- Trained and compared logistic regression, support vector machine, random forest, and gradient boosting classifiers to identify discriminative signatures of Hawaiian mantle compositions.
- Assessed trace-element predictors as lower-cost alternatives to isotope-based measurements, connecting model outputs to the composition and evolution of Earth's mantle.